

Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech.
Code: 101306

Subject: Introduction to Civil Engineering

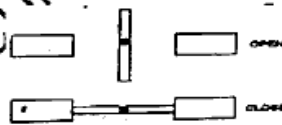
Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer/ fill in the blanks of the following (Any seven only): [2 x 7 = 14]

- (a) Pneumatic Structure contains
(i) Metal (ii) Compressed air (iii) Wood (iv) Water
- (b) A building can be mainly divided into ----- components
(i) 1 (ii) 2 (iii) 4 (iv) 3
- (c) Additional wall used to resist lateral forces (wind forces, earthquakes forces)
(i) Knee wall (ii) Infill wall (iii) Shear wall (iv) Cavity wall
- (d) Machine foundation is subjected to
(i) Static & dynamic Loads (ii) Kinematic Loads
(iii) Both (i) & (ii) (iv) None of the above
- (e) Which IS code book guidelines for preparation of Mortar.
(i) IS 456-2000 (ii) IS 2250-1981
(iii) IS 3350-1981 (iv) None of the above
- (f) While storing, a gap of 0.3 meter is to be maintained between cement bag and wall.
- (g) The satellite Constellation of GPS consists of
(i) 4 satellites (ii) 6 satellites
(iii) 18 satellites (iv) 24 satellites
- (h) The figure below shows mechanism of



- (i) Sliding Door (ii) Spinning Door (iii) Swing Door (iv) Rotating door
- (i) Usually, speed of Escalator is
(i) 25-30 m/min (ii) 10-25 m/min
(iii) 40-50 m/min (iv) 30-45 m/min
- (j) What is the name of defect that will occur if paint is applied excessively thick?
(i) wrinkling (ii) Running
(iii) Flaking (iv) Grinning

- (a) List and explain, types of building classification as per national building code. [7]
- (b) What is an ASPECT in building planning? Explain PROSPECT principles in building planning. [7]

- Q.2**
- (a) With a neat sketch mention briefly about any two of modern surveying Instruments. [7]
 - (i) Electronic Distance Measurement
 - (ii) Total Station
 - (iii) Global Positioning
 - (b) Explain physical, mechanical properties & specifications of Indian standard bricks. [7]

- Q.4 (a) Difference between Pile and Pier Foundation. [7]
(b) Explain following Bearing capacity of soil. [7]
(i) Ultimate bearing capacity
(ii) Net safe bearing capacity
(iii) Gross safe bearing capacity
(iv) Net safe settlement pressure

- Q.5 (a) Explain the principles of surveying. [7]
(b) What are different types of **Design Loads** in designing any structure. Also discuss about load combination. [7]

- Q.6 (a) Mention name of two software from each domain and write their application in civil Engineering. [7]
(i) Drawing & drafting software
(ii) Design Software
(iii) Detailing Software
(iv) Project Management Software
(b) With a neat sketch and example (of Existing power plants in India), explain hydro power plant and thermal power plant working. [7]

- Q.7 (a) Name typical ports in East and west coastal part of India and discuss their economical Important. Also list name of structures available at ports & harbour. [7]
(b) Describe hydrological cycle with neat sketch and list various practical application of hydrology. [7]

- Q.8 (a) Write short note on any two basic infrastructure services? [7]
(i) Air conditioning & purposes
(ii) Fire protection and materials
(iii) Ventilation, necessity and functional requirements.
(iv) Lifts & Escalators
(b) What are the open space requirements as per National Building Code norms? [7]

- Q.9 Explain the following materials and list their various uses in building construction: [7x2=14]
(a) Plain concrete
(b) Reinforced cement Concrete <https://www.akubihar.com>
(c) Prestressed concrete
(d) Ready mix concrete
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